

REMOTE MULTI-PERSON BLOOD PRESSURE MONITORING USING MMWAVE RADAR

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ABSTRACT

Radar technology provides reliable blood pressure (BP) monitoring through advanced contactless sensing. However, most existing radar-based techniques are constrained to single-target scenarios, presenting challenges in accurately estimating BP in multi-target scenarios due to unreliable physiological signal extraction and random body movements. We propose a Bayesian-driven variational mode decomposition (BD-VMD) algorithm for accurate BP estimation via a millimeter-wave (mmWave) frequency-modulated continuous wave (FMCW) radar. The algorithm discriminates the phase signal (corresponding to noise and other physiological information) from that of the heartbeat activity via Bayesian optimization, ensuring a balance between reconstruction error and bandwidth constraints. A biological model is then employed for BP estimation. Experimental validation demonstrates the effectiveness and robustness of the proposed algorithm across diverse multi-person environments.

Index Terms— Blood pressure (BP) monitoring, Bayesian-driven, millimeterwave (mmWave), frequency-modulated continuous wave (FMCW)

1. INTRODUCTION

Blood pressure (BP) serves as an important indicator of cardiovascular health. Conventional measurement methods, such as mercury sphygmomanometers or wearable devices [1], require direct skin contact, which may cause discomfort with repeated use, particularly for patients with skin conditions or burns [2]. In recent years, radar technology has been explored for non-contact BP monitoring using both continuous wave (CW) [3] and frequency-modulated continuous wave (FMCW) radars [4]. These solutions estimate BP by analyzing arterial pulse transit signals from the waist or neck. However, these approaches are limited by specific radar placement requirements, short operating distances (approximately 30 cm), and the need for subjects to remain stationary [5].

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To this end, alternative methods have been proposed to exploit chest surface motion as a BP surrogate or pulse wave separation via variational mode decomposition (VMD) [6] to reduce interference. Most existing approaches also focus on single-person scenarios and chest motion rather than the precise phase signals of heartbeats, making multi-person BP estimation challenging due to signal attenuation and inter-subject interference [7]. While multi-person heart rate (HR) monitoring using the FMCW radar has been explored [8], this approach applies filtering techniques and, therefore, suffers from poor performance in the presence of random body motion [7]. Moreover, BP estimation requires accurate reconstruction of fine-grained heartbeat waveforms since the presence of minor distortions, noise, or mode mixing can significantly impair BP inference, despite seemingly accurate HR values.

To address the aforementioned limitations, we propose a Bayesian-driven variational mode decomposition (BD-VMD) algorithm for BP monitoring in multi-person scenarios. Unlike conventional VMD, which suffers from mode mixing and relies on manually selected parameters [9], BD-VMD leverages Bayesian optimization to determine the optimal number of modes and the penalty coefficient. This adaptive decomposition improves the separation of physiological components in the presence of random body movements, allowing the extraction of high-fidelity heartbeat signals with preserved waveform integrity that are essential for accurate BP mapping. Experiment results across different distances, environmental conditions (indoor and outdoor), and number of participants demonstrate the higher accuracy and robustness of the proposed BD-VMD algorithm over existing radar-based BP estimation methods.

2. SIGNAL MODEL AND HUMAN TARGET LOCALIZATION

2.1. Signal Model

Figure 1 illustrates a representative scenario for multi-person blood pressure monitoring using an FMCW radar. The chirp signal is transmitted via the transmit (Tx) antenna with a linearly increasing frequency f_0 to f_t within the chirp dura-

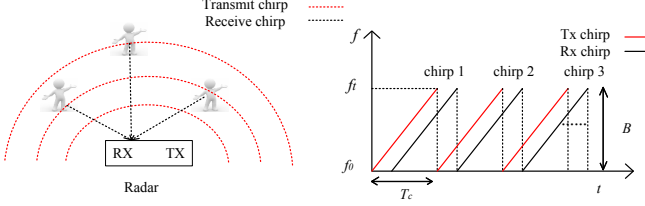


Fig. 1. FMCW radar system for multi-person BP monitoring.

tion T_c . Let P denote the number of people within the radar field of view, indexed by $p = 1, \dots, P$, where the p -th person is located at a radial distance $R_{p,0}$ and direction θ_p . We assume the radar system is configured as a uniform linear array (ULA) with M receive (Rx) antennas, indexed by $m = 1, \dots, M$. The intermediate frequency (IF) signal at the m th antenna is given by

$$x_m(t) = \sum_{p=1}^P \alpha_p e^{j2\pi \left(f_{b,p}t + \frac{2R_{p,0}}{\lambda} + \frac{u_n}{\lambda} \sin(\theta_p) \right)} + n(t), \quad (1)$$

where $j = \sqrt{-1}$, α_p denotes the reflection coefficient of the p th person, $f_{b,p} = \frac{2BR_{p,0}}{cT_c}$ is the beat frequency associated with the p th person, λ is the wavelength, and $n(t)$ is the noise component. Here, B denotes the bandwidth, c is the speed of light, and $R_p(t) = R_{p,0} + \Delta R_p(t)$, where $\Delta R_p(t)$ is the change in radial distance attributed to the cardiopulmonary activity of the p th person.

After analog-to-digital (ADC) conversion, the discretized IF signal at the m th antenna can then be expressed as

$$x_m(k, l) = \sum_{p=1}^P \alpha_p e^{j2\pi \left[f_{b,p}k + \frac{2R_{p,0}}{\lambda} + \frac{2\Delta R_p(l)}{\lambda} + \frac{u_n}{\lambda} \sin(\theta_p) \right]} + n(k, l), \quad (2)$$

where $k = 0, \dots, K-1$ is the fast-time sample index with K being the number of samples per chirp, $l = 0, \dots, L-1$ is the slow-time index with L being the number of chirps, and $n(k, l)$ is the corresponding noise component. In practice, the signal is sampled across multiple chirps during ADC conversion. The small chest displacement of the person causes the phase of the IF signal to vary along the slow-time dimension such that [7]

$$\phi_{m,p}(l) = \phi_{m,0} + \frac{4\pi\Delta R_p(l)}{\lambda}, \quad (3)$$

where $\phi_{m,0} = \frac{2\pi}{\lambda} [2R_{p,0} + u_n \sin(\theta_p)]$ is time invariant given that the subject of interest stays at the nominal range $R_{p,0}$. It is evident that the second phase component $\frac{4\pi\Delta R_p(l)}{\lambda}$ contains the phase signal corresponding to the BP mapping.

2.2. Multiple target 2-D localization

To suppress the environmental clutter, the IF signal is averaged over the slow-time dimension such that

$$\tilde{x}_m(k, l) = x_m(k, l) - \frac{1}{L} \sum_{l=1}^L x_m(k, l). \quad (4)$$

To achieve the range information of each human subject, we first apply a K -point fast Fourier transform (FFT) on $\tilde{x}_m(k, l)$ across the fast time domain, i.e., $\tilde{\mathbf{x}}_m(l) = \mathcal{F}_{k \rightarrow f} \{ \tilde{x}_m(k, l) \}$, where $\mathcal{F}_{k \rightarrow f}$ denotes the FFT operator. The signal corresponding to the p th human target range bin across all antenna channels is, therefore, encapsulated within an $M \times 1$ vector [10, 11]

$$\mathbf{x}(l) = \mathbf{A}(\theta) s_p(l) + \mathbf{n}(l), \quad (5)$$

where $\mathbf{A}(\theta)$ denotes the array steering matrix comprising steering vectors $\mathbf{a}(\theta)$ towards θ , $s_p(l)$ is the signal corresponding to the p th subject, and $\mathbf{n}(l)$ denotes the noise vector.

With reference to (5), existing direction-of-arrival (DOA) estimation algorithms such as subspace-based methods [12] and sparse signal representation methods [13] can be employed to determine the spatial directions of individuals. As opposed to sparse systems [14–16] and since the number of targets is generally unknown in practice, we utilize the sparse signal representation method [13] to determine the direction information of the target subjects. This is achieved by solving

$$\min_{\mathbf{y}(l)} \|\mathbf{x}(l) - \mathbf{D}\mathbf{y}(l)\|_2^2 + \lambda \|\mathbf{y}(l)\|_1, \quad (6)$$

where $\|\cdot\|_2$, $\|\cdot\|_1$, and λ denote, the ℓ_2 -norm, the ℓ_1 -norm, and the regularization parameter, respectively. Here, the matrix $\mathbf{D} = [\mathbf{a}(\theta_1), \dots, \mathbf{a}(\theta_Q)]$ denotes an $M \times Q$ overcomplete dictionary that is constructed by concatenating steering vectors toward directions $\{\theta_q\}_{q=1}^Q$, where $Q \gg M$ denotes the sampling grids that cover the potential spatial range of incident signals. The estimated DOA of the p th subject $\hat{\theta}_p$ is obtained by mapping $\mathbf{y}(\tilde{l})$ onto a predefined search grid. Combined with range estimates, these DOA values enable 2D localization, leading to accurate downstream BP measurement [13].

3. THE PROPOSED BP ESTIMATION PIPELINE

3.1. Target-Oriented Demodulation

We employ the MVDR beamformer [17] to enhance the target signal while suppressing directional interference such that the output is given by

$$z_p(l) = \left\{ \frac{\mathbf{R}_x^{-1} \mathbf{a}(\hat{\theta}_p)}{\mathbf{a}^H(\hat{\theta}_p) \mathbf{R}_x^{-1} \mathbf{a}(\hat{\theta}_p)} \right\}^H \mathbf{x}(l) \quad (7)$$

with $\mathbf{R}_x$ being the covariance matrix of $\mathbf{x}(l)$.

It is important to note that direct current (DC) offset remains the main source of phase distortion in BP estimation, introducing spectral leakage and drift. To address this, we employ nonlinear least squares (NLS) optimization [18] to remove DC bias while preserving amplitude and phase, effectively centering the signal within the complex plane. The objective function is given by

$$\min_{\Delta_x, \Delta_z, \bar{A}} \sum_l (|z_p(l) - (\Delta_x + j\Delta_z)| - \bar{A})^2, \quad (8)$$

where $\bar{A}$ denotes the average amplitude of $z_p(l)$, Δ_x and Δ_z represent the DC offset values for the real and imaginary parts, respectively. The phase signal corresponding to the p th person can then be extracted by $\phi_p(l) = \arctan\left(\frac{\text{Im}(\tilde{z}_p(l))}{\text{Re}(\tilde{z}_p(l))}\right)$, where $\text{Re}(\cdot)$ and $\text{Im}(\cdot)$ denote real and imaginary operators, respectively, and $\tilde{z}_p(l)$ denotes the signal after DC offset. The unwrapped phase signal $\phi_p^{(u)}(l)$, which is used for subsequent processing, is obtained recursively by compensating for phase wrapping between adjacent samples.

3.2. Heartbeat phase signal extraction via BD-VMD

In conventional VMD, the signal is decomposed into N intrinsic mode functions (IMFs) $u_n(l)$, each representing a finite-bandwidth component [19]. The decomposition aims to minimize the total bandwidth across all IMFs while constraining their sum to reconstruct the original signal [20], i.e.,

$$\min_{\{u_n\}, \{\omega_n\}} \left\{ \left\| \sum_{n=1}^N \frac{\partial}{\partial l} \left[\delta(l) + \frac{j}{\pi t} * u_n(l) \cdot e^{-j\omega_n l} \right] \right\|_2^2 \right\}, \quad (9)$$

where ω_n represents the center frequency of each IMF, and $\delta(l)$ is the Dirac function. To extract heartbeat-related phase signals, we analyze the instantaneous phase variations represented by the unwrapped phase $\phi_p^{(u)}(l)$. The decomposition then approximates $\phi_p^{(u)}(l)$ as the sum of IMFs, i.e., $S_n(l) = \sum_{n=1}^N u_n(l)$.

However, VMD and its variants [21] suffer from rigid parameter tuning, reconstruction errors, and spectral overlap. To address this, we propose the BD-VMD algorithm, where Bayesian optimization is employed to extract BP by adaptively selecting the number of modes N and penalty coefficient α using the cost function

$$J(\alpha, N) = \frac{\gamma}{L} \sum_{l=1}^L \left(\phi_p^{(u)}(l) - S_n(l) \right)^2 + \alpha \sum_{n=1}^N \left\| \frac{\partial U_n(f)}{\partial f} \right\|_2. \quad (10)$$

This cost function balances reconstruction fidelity and spectral compactness with γ being the weighting parameter and $U_n(f)$ is the Fourier transform of the IMF $u_n(l)$. To this end, we propose a Bayesian optimization-based method to obtain the optimal decomposition parameters such that

$$(\alpha^*, N^*) = \arg \min_{\alpha, N} J(\alpha, N). \quad (11)$$

With (α^*, N^*) , the proposed BD-VMD extracts the phase component within the heartbeat band, which is then reconstructed for analysis. Since random body movements and chest micro-displacements can mask the heartbeat signal with respiration or low-frequency motion, BD-VMD's optimized modal decomposition suppresses mode mixing and enhances signal separation, yielding a clearer phase component for accurate BP estimation.

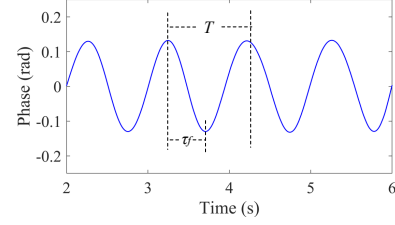


Fig. 2. Phase signal corresponding to the heartbeat activity.

3.3. BP Mapping

BP comprises systolic blood pressure (SBP), diastolic blood pressure (DBP), and pulse pressure (PP), with SBP and DBP commonly used to assess health. The heartbeat phase signal extracted via BD-VMD reflects arterial compliance, which depends on changes in arterial diameter and is primarily influenced by stroke volume (SV), the blood ejected per heartbeat [22]. An increase in SV prolongs the ventricular systolic period τ_f (which is mirrored in the phase signal as shown in Fig. 2) revealing the physiological basis of the BD-VMD algorithm. Given the established link between DBP and arterial compliance [23], DBP can be modeled as $\text{DBP} \propto e^{-\beta \tau_f}$, where β characterizes vascular properties. Pulse pressure, defined as the difference between SBP and DBP, depends on both SV and pulse wave velocity (PWV), with PWV reflecting arterial stiffness and its relation to τ_f such that

$$\text{PP} \propto \text{SV} \cdot (\text{PWV})^2 \propto \text{SV} \cdot \frac{\tau_f^2}{T}, \quad (12)$$

where T is the heartbeat period. The p -th person's heartbeat phase signal can then be mapped to $\text{DBP}^{(p)}$ and $\text{SBP}^{(p)}$ via

$$\text{DBP}^{(p)} = \text{DBP}_0^{(p)} \cdot e^{-\beta(\tau_f^{(p)} - \tau_{f_0}^{(p)})}, \quad (13)$$

$$\text{SBP}^{(p)} = \text{DBP}^{(p)} + \text{PP}_0^{(p)} \cdot \frac{\tau_f^2/T}{(\tau_{f_0}^{(p)})^2/T_0^{(p)}}, \quad (14)$$

where $\text{DBP}_0^{(p)}$, $\text{PP}_0^{(p)}$, $\tau_{f_0}^{(p)}$, and $T_0^{(p)}$ denote the initial DBP, SBP, τ_f , and T for the p th person, respectively. These values serve as algorithm initialization.

4. EXPERIMENT RESULTS

To evaluate the proposed BD-VMD algorithm, we collected data using the Texas Instruments AWR2243 mmWave radar [7], with Omron HEM-7361T BP monitors providing ground truth. Five experiments were conducted across varying target-to-radar distances, environments (indoor and outdoor), and participant numbers to assess accuracy and robustness. Performance was compared with the caPTT (a machine learning based method) [24], the approach in [25], the VMD algorithm [6], and the proposed BD-VMD algorithm.

We first evaluate the performance of various algorithms in outdoor scenarios with a single person sitting at different distances, as shown in Fig. 3(a). The mean absolute

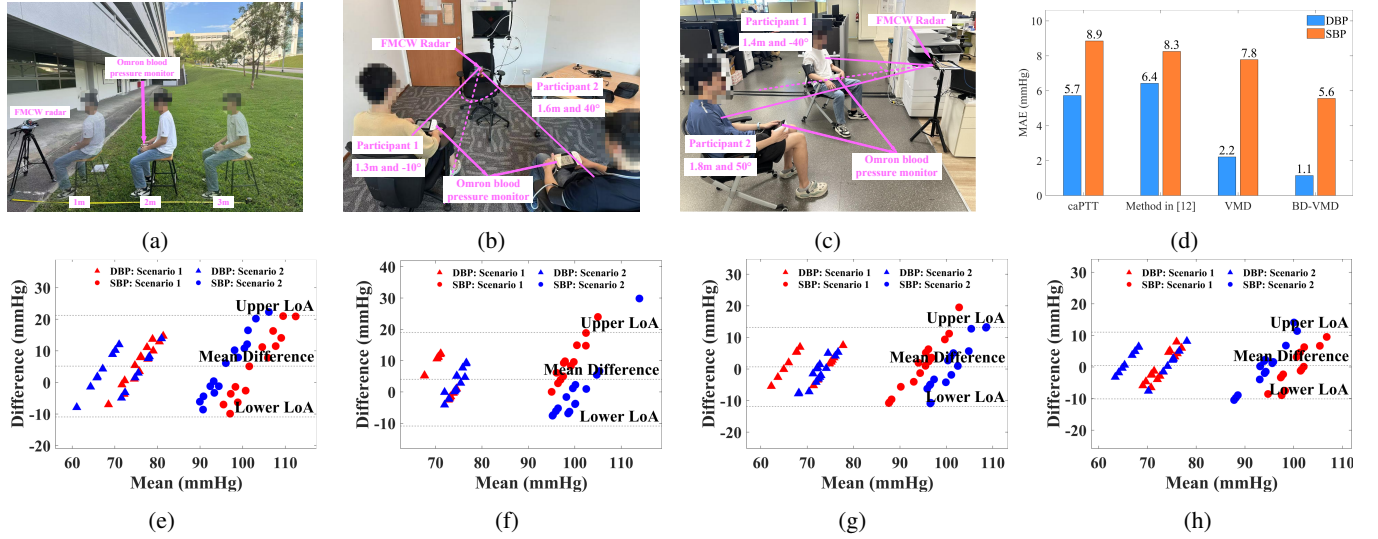


Fig. 3. BP estimation performance comparisons under different scenarios and algorithms. (a) Single person scenario under different ranges, (b) multi-people in a meeting room, (c) multi-people in a laboratory, (d) single person scenario's result, (e) caPTT method, (f) method described in [25], (g) VMD, (h) the proposed BD-VMD.

error (MAE) is employed as the evaluation metric and the aggregated estimation results are presented in Fig. 3(d). These results indicate that the caPTT method and the method in [25] exhibit limited accuracy in DBP estimation, with MAEs of 5.7 mmHg and 6.4 mmHg, respectively. In contrast, the proposed BD-VMD method significantly reduces the DBP estimation error to 1.1 mmHg. In terms of estimating the SBP, caPTT, the method in [25], and VMD yield MAEs of 8.9 mmHg, 8.3 mmHg, and 7.8 mmHg, respectively, whereas BD-VMD achieves the lowest MAE of 5.6 mmHg. Overall, BD-VMD consistently outperforms the three baseline methods, demonstrating high accuracy and robustness in outdoor scenarios.

To assess the performance of BD-VMD in multi-person scenarios, we conducted two experiments involving two participants positioned at different ranges and angles within a meeting room and a laboratory setting, as illustrated in Fig. 3(b) and (c). The results are visualized using Bland-Altman plots shown in Fig. 3 (e)-(h), where the ordinate represents the mean of the ground truth and the estimated values, while the abscissa shows their differences. These plots highlight the error distribution and limits of agreement (LoA) between methods. Notably, BD-VMD exhibits a mean difference closest to zero among all compared methods, indicating the highest agreement with the ground truth and improved BP estimation accuracy. Furthermore, the baseline algorithms exhibit wider LoA compared to BD-VMD, indicating greater variability in their BP estimations. To further evaluate the estimation performance, we analyze the root mean square error (RMSE) and error distribution intervals, summarized in Table 1. BD-VMD achieved the lowest RMSE of 5.37 mmHg among all methods, underscoring its

Table 1. Performance Comparison

	caPTT	Method in [25]	VMD	BD-VMD
Mean difference	9.64	4.06	0.65	0.47
Upper LoA	25.73	19.02	13.17	11.05
Lower LoA	-6.45	-10.89	-11.87	-10.11
RMSE	9.64	8.58	6.47	5.37
Error within $[-5, 5]$ mmHg	34.34%	45.76%	54.59%	63.93%
Error within $[-10, 10]$ mmHg	68.85%	81.36%	85.94%	95.08%

superior accuracy. In addition, 63.93% of BP estimates fall within the $[-5, 5]$ mmHg range, and 95.08% within the $[-10, 10]$ mmHg range, both being the highest among the compared methods. These results highlight the strong reliability and consistency of BD-VMD. They also validate the robustness in delivering stable and accurate BP measurements in complex multi-person scenarios.

5. CONCLUSION

We proposed the BD-VMD algorithm, designed to enhance BP estimation from radar signals in multi-person scenarios by accurately extracting phase related to heartbeat. BD-VMD adaptively optimizes the number of modes and bandwidth control parameters, ensuring a balance between signal reconstruction accuracy and modal bandwidth constraints, which enhances decomposition precision and robustness. Compared to previous methods, BD-VMD demonstrates superior performance in both error distribution and agreement range, leading to high-accuracy, non-contact BP estimation in multi-person scenarios. These results highlight the robustness of BD-VMD, and future work will focus on extending the proposed method to more complex real-world scenarios.

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